

[6578]-27

S.E. (Information Technology) (Insem.)
DISCRETE MATHEMATICS
(2019 Pattern) (Semester - III) (214441)

Time : 1 Hour]**[Max. Marks : 30****Instructions to the candidates:**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) In a group of 100 students, 72 students can speak English and 43 students can speak Hindi. Each student speaks at least 1 language. Based on this data, answer the following questions : **[5]**

- i) Find the number of students who can speak English only.
- ii) Find the number of students who can speak Hindi only.
- iii) Find the number of students who can speak both English and Hindi.

b) Define tautology. **[5]**

Prove by constructing the truth table:

$$P \rightarrow (Q \vee R) \equiv (P \rightarrow Q) \vee (P \rightarrow R)$$

c) Translate the following statements into logical notation, using the following symbols: **[5]**

S = "The Stromboli is hot."

L = "The lasagne is cold."

P = "The pizza will be delivered."

- i) "The stromboli is hot and the pizza will not be delivered."
- ii) "If the lasagne is cold, then the pizza will be delivered."
- iii) "Either the lasagne is cold or the pizza won't be delivered."
- iv) "If the pizza won't be delivered, then both the stromboli is hot and the lasagne is cold."
- v) "The lasagne isn't cold if and only if the stromboli isn't hot."

OR

Q2) a) For a given multiset find the following operations **[5]**

$$A = [a, a, b, c, d, d, d, e] \quad B = [a, b, d, f, g]$$

$$C = [b, c, e, e, g, h, h] \quad D = [a, d, d, e, f, fg, h]$$

Find

- i) $A \cup B$
 - ii) $C \cap B$
 - iii) $A - D$
 - iv) $B + C$
 - v) $A \cap B$
- b) Prove using mathematical induction “ $6^n - 1$ is multiple of 5” **[5]**
- c) i) Write the solution set of the equation $x^2 - 4 = 0$ in roster form.
- ii) Write the set $A = \{1, 4, 9, 16, 25, \dots\}$ in set-builder form.
- iii) Write the set $B = \{\dots, -8, -4, 0, 4, 8, \dots\}$ in set-builder form.

[5]

Q3) a) In how many ways can 7 men and 7 women be seated in a line so that no two Women sit together? **[5]**

b) How many 3-letter words with or without meaning can be formed out of the letters of LOGARITHMS if repetition is not allowed? **[5]**

c) A man is informed that when a pair of dice were rolled, the result was seven. How much information is there in this message? **[5]**

OR

Q4) a) Given that you drew a red card, what's the probability that it's an ace? **[5]**

b) How many bit strings of length 7 either start with 1 bit or end with 2 bits 00? **[4]**

c) Find the number of permutations that can be made out of the letters **[6]**

i) COMPUTER

ii) DIVIDEND

iii) COMMISSION

